



L5.4: What is a linear mapping - Part 2



Transcript Language

English



Transcriber's Name: Crescendo Transcriptions Pvt Ltd

Mathematics of Data Science 2 Professor Sarang Sane

Department of Mathematics Indian Institute of Technology, Madras

What is a linear mapping Part 2

Let us move on to a slightly more elaborate example. So, we will use the same data that

we have before, we have shop A. But now in this locality Malgudi, on this town Malgudi,



we have the original shop A, and we have two other shops B and C as well, and the prices

of rice, dal and oil in each of the shops is given in this table.

So, in Shop A we already have seen that it is 45 kgs, 45 rupees per kg for rice, 125

rupees per kg for a dal and I think 150 rupees per liter for oil. And for Shop B we have

40, 120 and 170 respectively and Shop C we have 50, 130 and 160 respectively.

Now, someone asked the caterer why do you always buy from shop A? So, is there a good

way of deciding, which shop to buy your groceries from? So, in order to do this, what do we

have to do, we have to check out the cost for buying X_1 kgs of rice, X_2 kgs of dal and

X_3 liters of oil from each of the shops, then compare them and see which one is less, and

then we would go there. Of course, let us assume that all of them have the same quality

and there is no difference as such.

So, let us do that. So, we have seen already that the expression for shop A is a linear

combination, and we can, in fact view it as a function. That is what we saw in a few slides

ago, and we can also think of it as matrix multiplication. Just to remind you, we had

this thing called c_A , which was the function, which was this linear combination 45×1 plus

125×2 plus 150×3 three, and which we can also write as a matrix multiplication.

So similarly, for shops B and C, we can write the cost function c_B and c_C . So, the shop

is capital C and the cost is little c. And expressions and matrix forms are as below.

So, c_B is the linear combination 40×1 plus 120×2 plus 170×3 and that is the corresponding

matrix multiplication. And for shop C, we have the function c_C , which is 50×1 plus 130×2

plus 160×3 and the corresponding matrix multiplication.

So, to compare these expressions, let us go back for a second and check them out. We had

45×1 plus 125×2 plus 150×3 , 40×1 plus 120×2 plus 170×3 . So, rice and dal are cheaper in

B than in a, but oil is more expensive. And then we have 50×1 plus 130×2 plus 160×3 . So,

rice and dal in C are more expensive than in shop B, but oil is less expensive.

However, when you compare it with shop A, all three are more expensive for shop C. So,

you would, assuming everything else is the same you would always prefer shop C, shop

A over shop C. So that is what we see here.

The third expression always yields larger values than the first one. Of course, we are

talking about commodities, so we think of them as buying the, I mean, buying commodities

from a grocery shop. So x_1 , x_2 , x_3 are always I think of them as positive numbers, but we

do not know how to compare between the second shop that is shop B and shops A and C.

So, to compare this, we would want to compare these cost functions. So, the idea you would

I mean, the natural way of doing this would be to treat them together c_A , c_B and c_C we

could think of them as a vector. So, you look at this vector, c_A , c_B , c_C and then we can

think of this actually as a function. Because each of these, after all are functions from

$\mathbb{R}^3$ to $\mathbb{R}$, so this vector can be thought of as an element of $\mathbb{R}^3$, it is a point of $\mathbb{R}^3$.

So, we can think of the cost function which is C of X_1, X_2, X_3 is c_A of X_1, X_2, X_3 comma

c_B of X_1, X_2, X_3 comma c_C of X_1, X_2, X_3 , I will just put this bracket here. And just

to recall each of these are given by linear expressions, linear combinations $45x_1$ plus

$125x_2$ plus $150x_3$ and then $40x_1$ plus $120x_2$ plus $170x_3$ and then $50x_1$ plus $130x_2$ plus $160x_3$.

So, this is the function, the cost function, wherein

the first coordinate of the cost function tells you the cost in shop A, the second coordinate

tells you in shop B, and the third coordinate tells you in shop C. And now, when you are

given X_1, X_2, X_3 you put it into this cost function, look at these three numbers that

it produces and then you choose the smallest one, you can just read them off. So, this

would be a good way of trying to compare. Of course, there are many other reasons to

look at the cost function.

For example, we can treat this as matrix multiplication. Because after all each of these are given

by matrix multiplication, where we have a column vector X_1, X_2, X_3 on the right and

a row vector corresponding to the cost of individual items on the left. So, you can

put these three row vectors together and get this matrix.

So, C of X_1, X_2, X_3 has a matrix form where the matrix is 45, 125, 150. This was the prices

for shop A, 40, 120, 170, which are the prices for shop B and for 50, 130, 160, which are

the costs for shop C. So, this is exactly what your table was the cost table, a few

slides ago. And you can see by just multiplying out that this is indeed what the cost vector

is.

So, just as an example, the cost of buying 2 kgs of rice, 1 kg of dal and 2 liters of

oil are given by the cost vector, you put X_1 is 2, X_2 is 1 and X_3 is 2 and we will get

515, 540 and 550, if hopefully I have done this computation correctly.

So as in the case of the function c_A , the property of linearity of costs can now be

extracted from the matrix form for the cost function c . So, let us first of all understand

this statement. What do we mean by the statement, the statement means that if you buy, so suppose

again, we had a situation like Monday and Tuesday, and then we had a say we wanted to

know what the costs for each shop on Wednesday are, so I would be evaluating this on half

times the rice on Monday, plus 5 by 4 times rice on Tuesday comma halftimes rice, dal

excuse me, dal on Monday plus 5 by 4 times dal on Tuesday, and then half times oil on

Monday plus 5 by 4 times oil on Tuesday.

You would be evaluating it on this, which means you have this matrix. Well, let me come

to the matrix in a second, which means basically you are doing c_A of the same thing, c_B of

the same thing and c_C of the same thing, but we saw that c_A is basically after we finished

the computations the c_A of this is exactly half times c_A on Monday. So, I will just call

it Monday.

By Monday, what do I mean? I mean rice on Monday, the vector rice on Monday, dal on

Monday comma oil on Monday, plus 5 by 4 times cA on Tuesday, which means by T I mean the

vector rice on Tuesday, dal on Tuesday, oil on Tuesday. And then I can, the same way as

I did for cA the same. There is nothing special about the shop A. So, the same thing works

for cB as well and for cC as well.

So, I have this very nice expression, and then half times cC of Monday plus 5 by 4 times

cC on Tuesday. And so, I can effectively compute this by computing everything on Monday and

Tuesday, but it gets even better because we know how to do matrix. We know how to do,

how to add and subtract and scalar multiply things in $\mathbb{R}^3$.

So, because we can do that, we can write this in even better as half times the cost vector

evaluated on the commodities for Monday, plus 5 by 4 times the cost vector evaluated on

the vector for Tuesday. I can separate these into two different vectors, and then I can

pull out the scalars, I will encourage you to do that. I will just, just in case it was

confusing what is M. By M I meant rice on Monday then dal on Monday, oil on Monday.

And by T I meant rice on Tuesday, dal on Tuesday, oil on Tuesday.

So, notice, I do not even remember what the quantity is there. I did not need that anywhere.

So, I can instead replace this rice and dal and so on by variables. I could have called

them X_1 , X_2 , X_3 and Y_1 , Y_2 , Y_3 and the whole thing would have worked out in the same way.

So effectively, what we are saying by linearity is that if you have α times X_1 , X_2 , X_3

plus Y_1 , Y_2 , Y_3 this is going to be just α times C of X_1 , X_2 , X_3 plus C of Y_1 , Y_2 , Y_3 .

That is what we mean by linearity.

This is a very important property, and this is very special to the kind of functions that

we are dealing with, they are linear functions. So that is what this video was about. Remember,

linear mapping. These are examples of linear mappings. Now, the claim is this can be, we

could have done this in one shot by using the matrix. How is that? Because remember

that, so let me just take here.

So, I can write this as the matrix times the vector X_1, X_2, X_3 plus the

vector Y_1, Y_2, Y_3 . But we know that matrix multiplication commutes with the addition.

So, I can instead write this as the matrix, the cost matrix that we had in the earlier

slide, times X_1, X_2, X_3 , and I can pull the α out because of constants remember come

out plus the same matrix times Y_1, Y_2, Y_3 .

And then, of course, this I know is α times C of X_1, X_2, X_3 . I mean it is the same

as the expression on top. It is equal to this expression. This this part is here and the

second part is corresponding to C of Y_1, Y_2, Y_3 . So, this is what we mean by a linearity.

So now, we can talk in general, what is a linear mapping? So, a linear mapping f from

$\mathbb{R}^n$ to $\mathbb{R}^m$. And now, we are not doing $\mathbb{R}^3$ or $\mathbb{R}$ we will just do it for any arbitrary n and

m . mean something of this form where you have f of x_1, x_2, x_n is summation $a_{1j}x_j$ summation

$a_{2j}x_j$ summation $a_{mj}x_j$, what are these a_{ij} s, they are real numbers.

So, basically a linear mapping can be thought of as a combination or a, sorry, a collection

of linear combinations. Except you arrange them in a vector, so it gives you a function.

So linear mapping is a function, which has this form. So just to be clear, if you take

something like f of x is x squared, that is not a linear mapping or f of x is logarithm

of x that is not a linear mapping or f of x is e to the power x that is not a linear

mapping. So, it should be linear, meaning it has this is very nice form.

So, we can of course write this as matrix multiplication, because notice that the right-hand

side, if we think of it, instead of a row vector, if we think of it as a column vector

then we can obtain it as multiplying A times x . What is A ? A is exactly the matrix given

by putting these coefficients into the appropriate places. So, the matrix the ij th entry of A

is a_{ij} where a_{ij} is the corresponding coefficient above.

So finally, let us talk about linearity of linear mappings. This is exactly what we did

for the cost vector. So, it follows that a linear mapping satisfies linearity, which

means f of X_1 plus Cy_1 comma X_2 plus Cy_2 , comma X_n plus Cy_n is equal to f of $X_1, X_2,$

X_n plus C times f of Y_1, Y_2, Y_n .

Why is that? So, the reason is because I can write f of X_1 plus Cy_1 comma X_2 plus $Cy_2,$

X_n plus Cy_n as A times X_1 plus Cy_1, X_2 plus Cy_2, X_n plus Cy_n . So, here of course, I am,

when I write it as A times just, I get a column letter, but you think of it as a row vector.

So, when I say equal, I mean, you take the row vector corresponding to this column vector.

So, the second I mean the column vector over there X_1 plus Cy_1 I can break up into two

separate vectors along with a constant. And then because matrix multiplication is commutes

with or distributes over addition, we get this, which is exactly f of X_1 , X_2 , X_n plus

C times f of Y_1 , Y_2 , Y_n . You could do this directly also, by the way, and I will encourage

you to check this from the expression for the linear combinations.

So, just to recall what everything that we have done in this video. We saw the examples

of grocery shops, first of a single grocery shop and the corresponding cost vector, and

then of a bunch of grocery shops. And we saw that it was useful to arrange this as a function.

We can treat the cost as a function, it is a linear combination, and that is the expression

for the function, and it satisfies what is called linearity.

And then if we have several costs, which is what we had four different jobs, we arrange

them into a cost function of which is a function of three variables. Taking values, depending

on how many shops there are. So, it could be in, so we had three shops, so it took values

in R^3 so it is a function from R^3 to R^3 , but, each coordinate of the expression that we

had was a linear combination of the X_1, X_2, X_n of X_1, X_2, X_3 . So, such a thing is exactly

what is called a linear mapping. And then we observed that linear mappings are satisfied

linearity. Thank you.